

- ✓ The assignment must be submitted within the deadline. **Late submissions will be penalized.**
- ✓ Submission of softcopy of the assignment is mandatory.
- ✓ Submit your assignment through this link - <https://forms.gle/99AfoEmZ2E6R1AeG8>

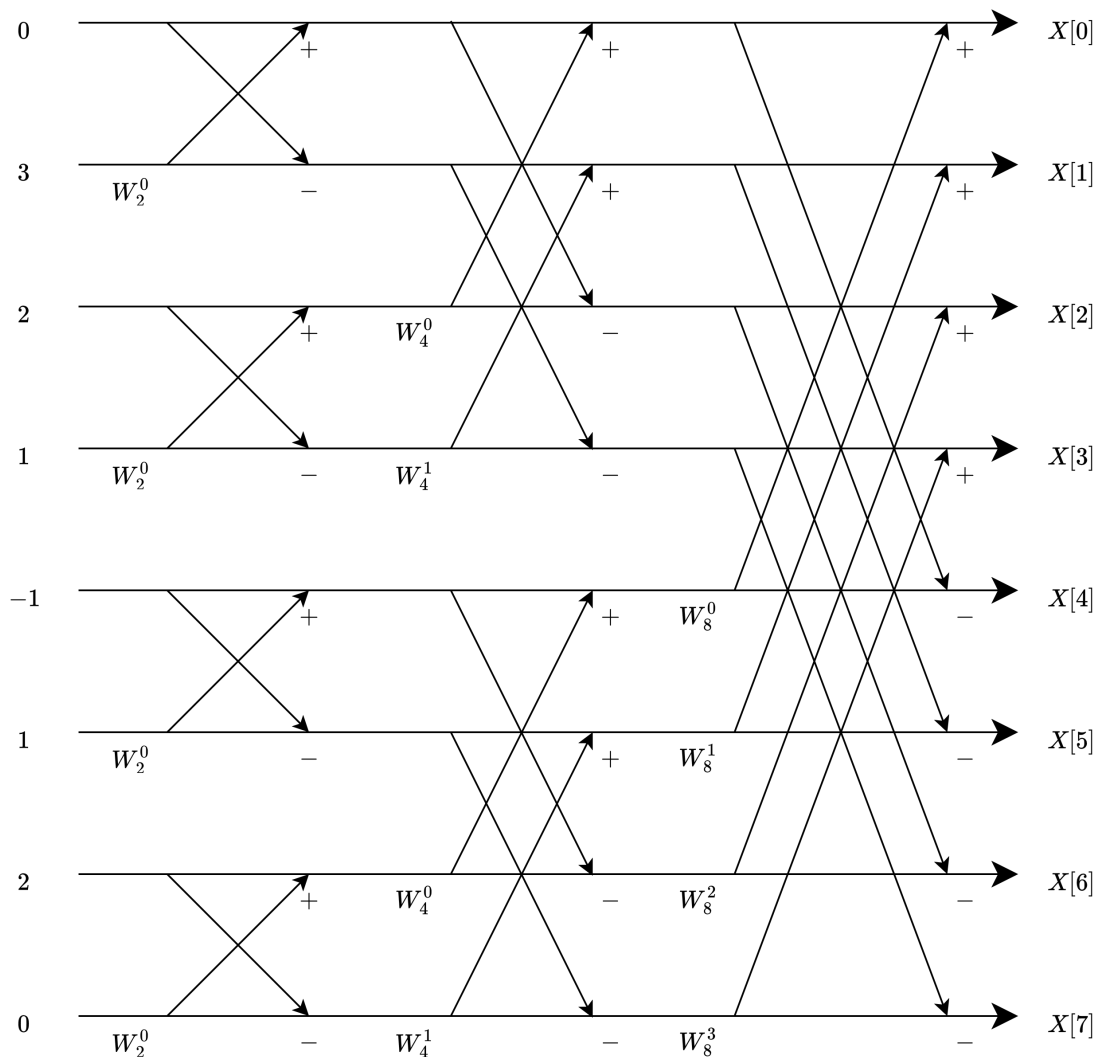
■ Question 1 of 4 [8 marks]

Draw the butterfly diagram for applying FFT on $x[n] = \{\underline{1}, -1, 2, 3, 1\}$. The underlined value marks the 0th index. You have to create a general butterfly diagram suitable to take $x[n]$ as an input and map the signal values on the input side. You do not have to calculate the outputs (neither final nor intermediate).

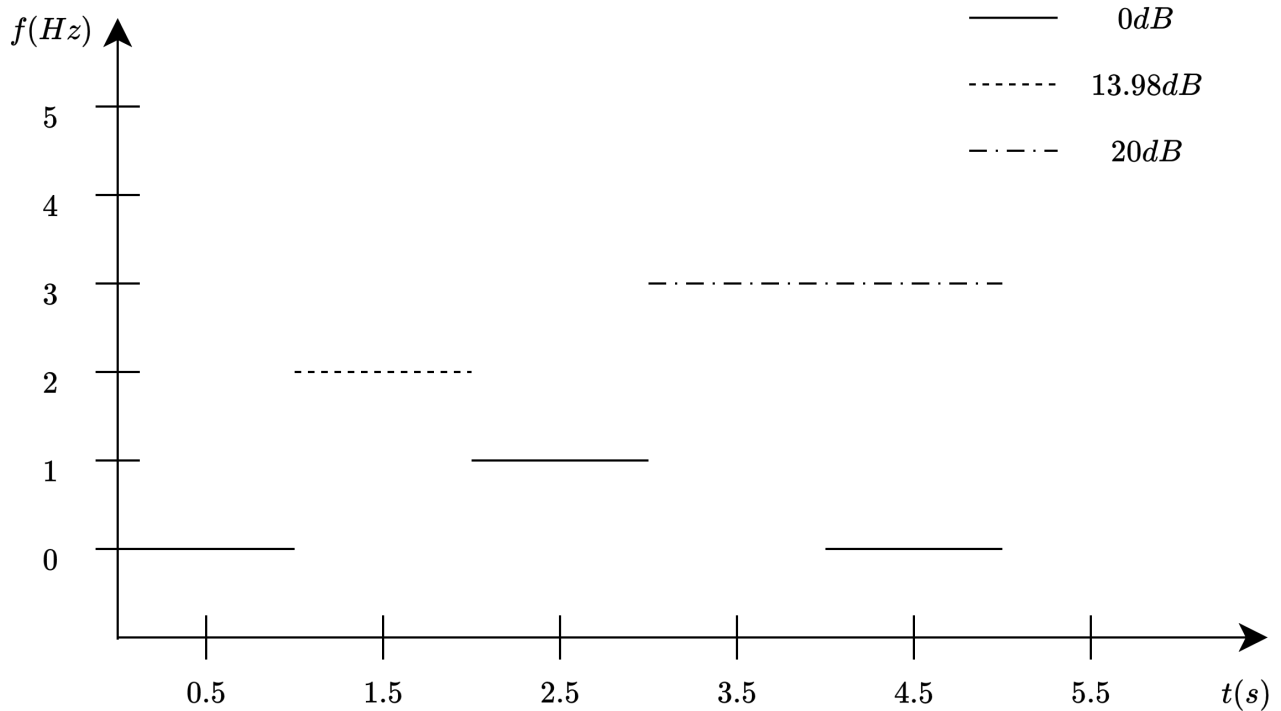
■ Question 2 of 4 [6 marks]

A butterfly diagram for 8 point FFT is given below. Find the input signal, $x[n]$.

NB - Learn the mapping and don't memorize or else you will get confused in the finals.



■ Question 3 of 4 [15 marks]



The spectrogram has been prepared using a centered window of window length 1 second with no overlap. Assume no window contamination. The $||X(f, t)||$ is provided in decibel (dB). Decibel is a comparative unit of power where the conversion is given by the formula,

$$dB = 10 \cdot \log_{10} \frac{P}{P_{ref}} = 10 \cdot \log_{10} \frac{V^2}{V_{ref}^2} = 20 \cdot \log_{10} \frac{V}{V_{ref}}$$

For this spectrogram, $V_{ref} = 1V$. With these info, answer the following questions.

- [3 marks] If the frequency information is available up to 5 Hz as shown in the figure, what is the sampling frequency of the time domain signal, $x[n]$?
- [3 marks] If the time information is available up to 5 seconds only, what is the number of samples in the time domain signal, $x[n]$ within this duration? *You need the answer of (a) to determine this information.*
- [9 marks] Draw the approximate CT representation, $x(t)$ (envelope) of the signal from the spectrogram information. The amplitude of the signal would be in V . For fractional value of amplitudes, just round it to the closest integer for ease of drawing.

■ Question 4 of 4 [16 marks]

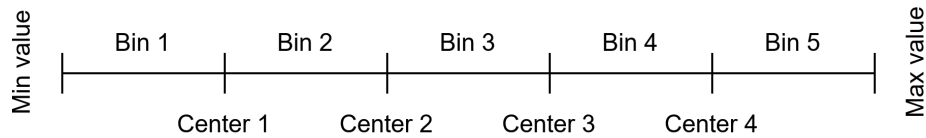
This is a guided math problem that will allow you to understand how you should solve problems with Mel spectrograms. Solve sequentially.

The following table provides 3 time bins of a spectrogram. The $||X(f, t)||$ are given in dB.

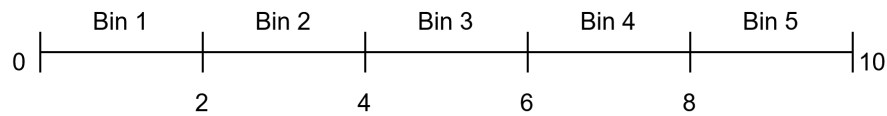
Frequency bins	t = 0 to 1 sec	t = 1 to 2 sec	t = 2 to 3 sec
0 kHz	5	3	4
1 kHz	25	27	21
2 kHz	22	25	17
3 kHz	23	24	20
4 kHz	19	21	19
5 kHz	16	15	13
6 kHz	12	14	11
7 kHz	7	11	10
8 kHz	6	9	6
9 kHz	5	7	3
10 kHz	3	7	2
11 kHz	2	6	0
12 kHz	-5	2	-5
13 kHz	-15	-5	-9
14 kHz	-20	-7	-17
15 kHz	-30	-9	-15
16 kHz	-50	-25	-23

Convert the max and min frequency bins to Mel at first. Round the values to integers for ease of calculation.

Divide the range between max and min Mel values into 4 centers $[m_1, m_2, m_3, m_4]$. 4 centers mean 5 uniform bins. Follow the figure provided below for reference,



Example -



$$\text{Bin length} = \frac{\text{max} - \text{min}}{n(\text{centers}) + 1}$$

With these info, now fill up the next table. For reference,

$$m = 2595 \cdot \log_{10}\left(1 + \frac{f}{700}\right)$$

$$\text{and, } f = 700 \cdot \left(10^{\frac{m}{2595}} - 1\right)$$

Mel center frequencies		Equivalent linear frequencies		Triangular filter bank limits		
m_{min}		f_{min}/f_0				
m_1		f_1		Filter 1	Lower limit, $F_{lower}^1 = f_0$	
					Upper limit, $F_{upper}^1 = f_2$	
m_2		f_2		Filter 2	Lower limit, $F_{lower}^2 = f_1$	
					Upper limit, $F_{upper}^2 = f_3$	
m_3		f_3		Filter 3	Lower limit, $F_{lower}^3 = f_2$	
					Upper limit, $F_{upper}^3 = f_4$	
m_4		f_4		Filter 4	Lower limit, $F_{lower}^4 = f_3$	
					Upper limit, $F_{upper}^4 = f_5$	
m_{max}		f_{max}/f_5				

At this point we have to figure out the frequency weights. Use this formula to get the weights,

$$W(f) = \begin{cases} 0, & f \leq F_{lower} \\ (f - F_{lower})/(F_{center} - F_{lower}), & F_{lower} \leq f \leq F_{center} \\ 1 - (f - F_{center})/(F_{upper} - F_{center}), & F_{center} \leq f \leq F_{upper} \\ 0, & f \geq F_{upper} \end{cases}$$

Do note that F and f both indicates frequencies in linear range. F demarcates frequency components associated with the filter banks. Also note that, according to the previous table,

$$F_{lower}^i = f_{i-1}$$

$$F_{center}^i = f_i$$

$$F_{upper}^i = f_{i+1}$$

Also note that the weight, $W(f)$ is basically $W^i(f)$ and is written without the superscript to avoid clutter.

	Triangular filter bank frequency weightage			
Frequency (kHz)	Filter 1	Filter 2	Filter 3	Filter 4
1				
2				
3				
4				
5				
6				
7				
8				
9				
10				
11				
12				
13				
14				
15				
16				

Finally, apply the weights to the time segments and derive the Mel spectrogram. Use the following formula to calculate each cell,

$$||X_{Mel}(m, t)|| = \sum_f W^m(f) \cdot ||X(f, t)||$$

Fill up the last table,

Mel centers		t = 0 to 1 sec	t = 1 to 2 sec	t = 2 to 3 sec
m_1				
m_2				
m_3				
m_4				

Note - In the final exam you won't be guided like this. Also the question will not be this big. You will be expected to do partial calculations from any of the segments.